

RONNIE KIYEGGA

+44 7775 633 768 • ronniekiyegga@hotmail.com • London, UK
[linkedin.com/in/ronniekiyegga](https://www.linkedin.com/in/ronniekiyegga) • github.com/BlissfulCoda • ronniekiyegga.com

SKILLS

Languages: TypeScript, JavaScript,
Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3
Backend: Node.js, Express.js, PostgreSQL, MongoDB
Infrastructure: Docker, GitHub Actions, AWS (S3), DigitalOcean

PROFESSIONAL EXPERIENCE

Software Engineer | School Of Research Science (EdTech Platform), Dubai, UAE Aug 2023 – Jan 2026

- Built an AI-powered browser-based programming IDE (**Next.js**, **TypeScript**, **PostgreSQL**, **ElevenLabs**) with an AST-based semantic analysis engine for real-time pseudocode validation used by ~1,000 students across multiple GCSE/A-Level computing classes.
- Engineered a client-side ML inference pipeline using **TensorFlow.js** for on-device image classification on memory-constrained iPad hardware, reducing classification latency from ~2s server inference to ~600ms on-device.
- Built a real-time analytics dashboard (**React**, **Chart.js**) aggregating student performance across departments, reducing end-of-term reporting workflows from ~20 hours of manual spreadsheet processing to under 1 hour.
- Introduced automated testing and CI pipelines (**Jest**, **Cypress**, **Playwright**, **GitHub Actions**) across 3 production applications, reducing QA cycle time from ~2 days to ~4 hours and enabling same-day releases.
- Translated GCSE/A-Level exam board specifications into automated grading workflows (Node.js, PostgreSQL), reducing average teacher marking time by ~3 hours per assessment cycle across 10+ classes.

Independent Software Engineer | London, UK Jun 2020 – Aug 2023

- Designed and delivered 8+ production full-stack platforms (Next.js, Node.js, Contentful CMS) for e-commerce and media clients, serving a combined audience of 50,000+ monthly visitors and generating recurring maintenance retainer contracts.
- Reduced media infrastructure costs by ~40% by replacing third-party video hosting with a custom **Node.js** and **FFmpeg** transcoding pipeline running on **AWS S3**.
- Designed and shipped **RESTful APIs** (Node.js, PostgreSQL) powering product catalogs and order workflows for 4+ e-commerce clients, handling 10K+ daily transactions and integrating with Contentful CMS for dynamic content management.

Software Engineer Internship | Adaptive Financial Consulting London, UK Aug 2019 – Dec 2019

- Built interactive React dashboards with D3.js timeline visualizations for 3 fintech product teams, enabling self-serve reporting for non-technical stakeholders and significantly reducing analyst data retrieval time.

PROJECTS

EduFeedbackPro – B2B SaaS AI-Analytics Platform

- Architected a multi-tier analytics pipeline (BigQuery → **NeonDB** → **MongoDB**) leveraging NeonDB as a **serverless caching layer** to reduce BigQuery query costs by ~60% and achieve sub-50ms **P95** API response times at scale.
- Integrated **ElevenLabs** speech APIs to enable teachers to query student analytics using voice commands, reducing time to identify at-risk students from several minutes of manual navigation to a single spoken query.
- Replaced **WebSocket** connections with **Server-Sent Events** (SSE) to eliminate persistent connection instability, reducing session failure rates by ~60% and improving platform reliability.

Ms Maryam's Math – B2C Tutoring Platform

- Architected a multi-role authentication system (JWT with automatic token rotation) supporting 3 distinct permission tiers, enabling secure concurrent access for 50+ active tutors and students on a live B2C platform.
- Configured CI pipeline (GitHub Actions) with automated test suites and blue-green deployments, reducing release cycle from ~20 minutes to under 3 minutes for a live tutoring platform serving paying customers.
- Secured API endpoints with Arcjet rate limiting and input validation across admin, tutor, and student roles, implementing defense-in-depth for a live multi-tenant tutoring platform.

EDUCATION

PGCE Computer Science • King's College London (*Postgraduate teaching qualification*)

Sep 2021 – Jun 2022

BSc (Hons) Computer Science • King's College London

Sep 2017 – Jun 2020